

Abstract

An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

9 I. SUMS OF SQUARES

10 The way to properly calculate the standard deviation or variances of groups is by using
11 Sums of Squares. Assuming you only have a set of means and standard deviations without
12 access to the original data, you need to be careful in how you compute a total standard
13 deviation, while the means are straightforward.

14 A. Means

15 The population mean is calculated as expected, simply by averaging the means. Let

$$\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]^T \quad (1)$$

16 be the individual data sets with means and standard deviations given by:

$$\boldsymbol{\mu} = [\mu_1, \mu_2, \dots, \mu_n]^T \quad (2)$$

$$\boldsymbol{\sigma} = [\sigma_1, \sigma_2, \dots, \sigma_n]^T. \quad (3)$$

17 Let $\mathbf{m}$ be a vector containing the number of observations in each dataset $\mathbf{x}_i$:

$$\mathbf{m} = [n_1, n_2, \dots, n_n]. \quad (4)$$

18 The mean of the entire population is the mean of the means:

$$\mu_{total} = \frac{1}{\sum n_i} \sum_i \mu_i. \quad (5)$$

19 B. Standard deviations

20 If the means are identical, the total variance is the geometric mean of the individual
21 variances. Otherwise, the Sum of Squares formulae are used. Let the sums of squares error,

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22 SSE, be given by

$$SSE = \sum_i n_i \sigma_i^2 \quad (6)$$

23 and the sum of squares of groups, SSG, be

$$SSG = \sum_i n_i (\mu_i - \mu_{total})^2. \quad (7)$$

24 The sum of squares total, SST, would normally be given by

$$SST = \sum_{i,j} (x_{ij} - \mu_{total})^2, \quad (8)$$

25 but since we don't have the individual x_{ij} values we use SSE and SSG:

$$SST = SSE + SSG. \quad (9)$$

26 The total variance, σ_{total}^2 , is finally given by

$$\sigma_{total}^2 = \frac{SSG + SSE}{\sum n_i}, \quad (10)$$

27 with the square root giving the standard deviation.

28 Note that if the datasets represent a sample rather than the entire population, equations
29 5, 6, and 10 should have n_i replaced with $n_i - 1$.